

Lab 5

Instructions for Lab Submission:

1. File Naming and Submission:
 - a. Submit your solution as a zip file named **lab5_<ERP>.zip**, where *<ERP>* is your ERP number. For example, if your ERP number is *12345*, the file should be named *lab5_12345.zip*.
 - b. Ensure that the zip file contains:
 - i. C++ source files named as **task<number>.cpp** (e.g., **task1.cpp**).
 - ii. No unnecessary files (such as executables, temporary files, or editor configs).
 - c. Do not submit **TEXT(.txt)** files.
2. Code Organization:
 - a. Write each task in its own **.cpp** file (as specified above).
 - b. Ensure that your code is modular and logically separated into functions when specified.
3. Code Neatness:
 - a. Ensure your code is well-formatted and easy to read. Use proper indentation and meaningful variable names.
 - b. Include appropriate comments to explain the logic where necessary (especially for functions and complex sections of the code).
 - c. **Add clear explanations where required (or specified)** to make the logic of your program easy to follow.
4. Code Compilation and Functionality:
 - a. Ensure your code compiles without errors or warnings. You should be able to compile and run your code on any system with a standard C++ compiler (e.g., GCC, Clang, MSVC).
 - b. Double-check that the functionality of the program meets the requirements outlined in the task.
5. Additional Instructions:
 - a. Do not use any libraries or features that have not been covered in the course material, unless specifically instructed.
 - b. Make sure your code adheres to the principles of good software development (e.g., modular functions, minimal repetition, and no hardcoding).
 - c. If you encounter any issues or require clarifications, reach out before the submission deadline.
6. Deadline:
 - a. Ensure that you submit your lab on time, before the deadline. Late submissions will not be accepted.
 - b. Double-check that the zip file is correctly named and contains all necessary files before submitting.
7. Plagiarism Policy:
 - a. The work submitted must be entirely your own. Any code or content copied from others (including online sources, classmates or AI) will result in an automatic zero for the task.

Lab Tasks

Task 1: Array Statistics

Write a program that asks the user to input ten numbers. The program should calculate and print the minimum, maximum, and average of the numbers. After that, print the numbers in reverse order of input.

```
fall25/Lab_Solutions/lab05 on  main [!?]
> g++ task1.cpp && ./a.out
Enter 10 numbers:
1 2 3 4 5 6 7 8 9 10
Minimum: 1
Maximum: 10
Average: 5.5
Numbers in reverse order:
10 9 8 7 6 5 4 3 2 1
```

Task 2: Multiplication Table

Write a program that generates an $n \times n$ multiplication table, where n is a user-defined size.

```
fall25/Lab_Solutions/lab05 on  main [!?]
> g++ task2.cpp && ./a.out
Enter the size of the multiplication table (n): 5
| 1 2 3 4 5
-----+-----
1 | 1 2 3 4 5
2 | 2 4 6 8 10
3 | 3 6 9 12 15
4 | 4 8 12 16 20
5 | 5 10 15 20 25
```

Task 3: Guessing Game

Write a C++ program that guesses a number in the range 0 to 100. The user tries to guess the number by input and the program gives the following feedback:

- 'low' if the guess is too low.
- 'high' if the guess is too high.

For generating a random number include this library `<cstdlib>`

```
I have chosen a number between 1 and 100. Try to guess it!
Enter your guess: 4
Too Low!
Enter your guess: 10
Too Low!
Enter your guess: 90
Too High!
Enter your guess: 50
Too Low!
Enter your guess: 60
Too Low!
Enter your guess: 80
Too High!
Enter your guess: 70
Too Low!
Enter your guess: 75
Too Low!
Enter your guess: 78
Too Low!
Enter your guess: 79
Congratulations! You guessed it.
```

Task 4: Count Occurrences

Given an array of integers and a target value, return how many times the target appears.